



Catarina Gamboa

Software Engineering Ph.D. Student

📍 Lisboa, Portugal

🏠 Portuguese nationality

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- Software verification researcher specializing in developer experience and verification systems with publications in top-tier A* conferences and 9+ workshop papers.
- ⚙️ Industry experience developing agentic workflows and static analysis tools at startup and enterprise environments.
- 📖 Cross-institutional teaching of 150+ students across CMU and University of Lisbon.

EDUCATION

2021–
NowA

Dual PhD in Software Engineering

CARNEGIE MELLON UNIVERSITY AND UNIVERSITY OF LISBON · Pittsburgh and Lisbon 🏛️

I am enrolled in the Dual PhD CMU-Portugal completing courses and research at both CMU and U. Lisbon. Currently, and I'm currently working on improving the usability of liquid types for reliable software, advised by Alcides Fonseca and Jonathan Aldrich.



2019–2022

Master Degree in Informatics Engineering - Software Engineering

FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DE LISBOA · Lisbon 🏛️

GPA of 18/20. The courses included Software Design (18/20), Compiling Techniques (19/20), Software Reliability (18/20), and Software Verification and Validation (18/20). Final Dissertation on "LiquidJava: Extending Java with Refinements" advised by Alcides Fonseca and Chris Timperley.



PROFESSIONAL EXPERIENCE

July-Oct
2025

Software Engineer PhD Intern

UBER, AMSTERDAM, NETHERLANDS

Designed and implemented agentic workflows using LLMs and custom MCP tools to automate Python library migrations for enterprise development teams.

May-Aug
2024

Software Engineer Research Intern

GITAR.CO, SAN MATEO, CA, USA

During this summer internship, I helped improve the static analysis tools across different programming languages, especially in the creation of def-use chains, and I contributed to the testing framework using fuzzing and applying the tool to open-source projects. I had the pleasure to work in the PL team with Raj Barik, Ameya Ketkar, Lazaro Clapp and Ali-Reza Adl-Tabatabai.

2020
-2025

Teaching Assistant at CMU and U. Lisbon

CARNEGIE MELLON UNIVERSITY AND UNIVERSITY OF LISBON

I was a T.A. for courses at both Carnegie Mellon and the University of Lisbon. At CMU I taught the course of API Design, working with Joshua Bloch on the best practices to design good APIs. During the course we focused on Java examples but explored other languages such as C and Rust. In Lisbon, I taught introductory courses on Programming to more than 100 undergraduate students of Information Technology, Mathematics, Physics and others.

PUBLICATIONS

FULL PAPERS

June 2025

Usability Barriers for Liquid Types

Catarina Gamboa, Abigail Reese, Alcides Fonseca, and Jonathan Aldrich. 📄

PLDI 2025 - Conference on Programming Language Design and Implementation, CORE A*. - We conducted an empirical research study analyzing usability challenges and users' barriers to using liquid type systems. We identified three main categories that can inform theoretical and practical advances in verification-based type system design.

May 2023

Usability-Oriented Design of Liquid Types for Java

Catarina Gamboa, Paulo Santos, Christopher S. Timperley, and Alcides Fonseca. 📄

ICSE 2023 - IEEE/ACM 45th International Conference on Software Engineering (ICSE), CORE A*. - We designed and evaluated LiquidJava, a human-centered extension of Java's type system using Liquid Types, we showed the improvement of bug detection and developer experience through participatory design and user studies.

SHORT PAPERS, POSTERS AND DEMOS

Dec 2024	Are large language models memorizing bug benchmarks? Daniel Ramos, Claudia Mamede, Kush Jain, Paulo Canelas, Catarina Gamboa , Claire Le Goues  Paper Accepted at LLM4Code'25 - This project examines data leakage in bug benchmarks for LLMs used for software engineering tasks, revealing memorization in some smaller models and stressing the need for better evaluation methods.
Oct 2023	Latte: Lightweight Aliasing Tracking for Java Conrad Zimmerman, Catarina Gamboa , Alcides Fonseca, Jonathan Aldrich  IWACO'23 and HATRA'23 @ SPLASH'23- We developed and formalized a lightweight aliasing and uniqueness tracking system that minimizes annotations while supporting common features in object-oriented languages such as Java.
Oct 2022	Poster - LiquidJava: Improving the Usability of Liquid Types for Reliable Software Catarina Gamboa , Alcides Fonseca, and Jonathan Aldrich  2022 CyLab Partners Conference, at Carnegie Mellon University, Pittsburgh - I presented an overview of my PhD projects.
Mar 2022	DEMO - Dive into LiquidJava - Extending Java with Liquid Types Catarina Gamboa , Paulo Santos, Christopher S. Timperley, and Alcides Fonseca.  Posters and Demonstrations track at <Programming> 2022 - I demonstrated how LiquidJava can be integrated in Visual Studio Code and used to verify Java programs that have Liquid Types or use libraries that are annotated with Liquid Types.
Jan 2022	Understandable and Useful Error Messages for Liquid Types Alcides Fonseca, Catarina Gamboa , João David, Guilherme Espada, Paulo Canelas  WITS'22 @ POPL'22 - We presented insights on the difficulties of handling error messages in languages with Liquid Types, drawing from experience with Aeon, LiquidJava, and MLVP, and connecting lessons to theorem provers like Coq and Lean.
Oct 2021	User-driven design and evaluation of Liquid Types in Java Catarina Gamboa , Paulo Santos, Christopher S. Timperley, and Alcides Fonseca  HATRA'21 @ SPLASH'21 - Presented the initial design of LiquidJava and validated it through developer surveys and a user study showing increased efficiency and adoption intent.
Sep 2021	LiquidJava: Adding Lightweight Verification to Java Catarina Gamboa , Paulo Santos, Christopher S. Timperley, and Alcides Fonseca  INForum 2021 - Informatics Symposium.
Nov 2020	Extending Java with Refinements Catarina Gamboa , Paulo Santos, Christopher S. Timperley, and Alcides Fonseca  PSSV'20 - Program Semantics, Specification and Verification: Theory and Applications, Student Paper

INVITED TALKS

Jan 2025	Talk @ PL@LX UNIVERSITY OF LISBON Shared my recently submitted work to a community of scholars in Lisbon that are interested in Programming Languages research.
Oct 2024	Guest Lecture @ Quality Assurance course CARNEGIE MELLON UNIVERSITY Chris Timperley invited me to give a guest lecture about static analysis, liquid types and LiquidJava in the course of Quality Assurance. The lecture counted with more than 50 students all enrolled in the Masters in Software Engineering at CMU.
Oct 2024	Talk @ RSS Meetup UNIVERSITY OF LISBON I was invited to share my internship experience during the summer of 2024, showcasing what I learned about how research can lead to successful startups and how to add a business perspective to research in programming languages.
Oct 2024	Talk @ Lasige Welcome Day - Into the Unknown: Journey from LASIGE to CMU and Silicon Valley Startups UNIVERSITY OF LISBON I was invited to share my experience of going to CMU and having a Summer internship in Silicon Valley, my approach was to give incentive to other students to aim high and not be afraid to go after international opportunities, and to bring to the University of Lisbon some of the practices I enjoyed having at CMU. This talk counted with Master and PhD students, as well as faculty members, with more the 100 people in the audience.
Dec 2022	Guest Lecture @ Secure Coding class CARNEGIE MELLON UNIVERSITY I was invited by Hanan Hibshi to give a talk about LiquidJava in the course of Secure Coding, 14735 during the Fall of 2022. This talk highlighted how type systems can be leveraged to improve code reliability and help developers find bugs at compile-time.

Jul 2022

Talk @ Lasige Forum - Science Communication and the Media

LASIGE, Faculdade de Ciências da Universidade de Lisboa

Presentation about the importance of Science Communication to broad audiences using media resources. Short version presented at the 1st LASIGE FORUM and a full-version presented in Talks@LASIGE.

CONFERENCE COMMITTEES

2023/2024

SPLASH'23 and '24 - Volunteer Co-Chair ([website](#))

CASCAIS, PORTUGAL AND PASADENA, USA

I was part of the organizing committee of SPLASH for two consecutive years, co-chairing the volunteering teams of around 50 volunteers. My responsibilities included managing schedules, assigning tasks, troubleshooting logistics, and fostering a welcoming and professional environment for more than 500 conference attendees.

2023

PLDI'23 - Artifact Evaluation Committee ([website](#))

ORLANDO, USA

I participated in the artifact evaluation committee of PLDI'23 and reviewed 3 different projects that accompanied the main conference accepted papers.

OUTREACH, LEADERSHIP AND TEAMWORK

May 2023

Student Volunteer at ICSE 2023, and 2024

MELBOURNE, AUSTRALIA, AND LISBON, PORTUGAL

I participated as a student volunteer in the organization of the conferences that counted with around 1500 participants.

2021-2023

Student Volunteer at POPL 2021, 2022 and 2023

I participated as a student volunteer in three consecutive years helping in the organization of the event in online, hybrid and in-person versions.

2021-2022

Co-organization of events at LASIGE

I joined the organization committee for the LASIGE Welcome Day 2021 and 2022, organized team building activities such as paintball and movie nights for all members of the research unit, joined the PhD student commission to help improve the well-being and connection among PhD students.

2019-2022

Board Member of the non-profit cultural organization Associação Alegres Olhares

I was an elected board member of the non-profit cultural association Alegres Olhares in the last two elections for the corporate entities. The association organizes and co-organizes cultural projects using creativity to take the traditional projects to our present day.

2010-now

Participation in Music and Theater Groups

I play the concert flute in several music groups including the Band of the Phillarminic Society of Olhalvo and the portuguese folk music group Noses com Vozes. I also participated in multiple theater groups such as the Academic Theater of the University of Lisbon (TUT) and the SCS Musical group at CMU.

AWARDS

Apr 2022

Huawei Scholarship

I was awarded with one of the 50 scholarships given in the scope of Huawei's program Seeds For The Future 2.0, in collaboration with .PT, to empower students in the STEM areas.

Sep 2021

Award for Best Student Poster

I received the award for best poster for the work on "LiquidJava: Adding Lightweight Verification to Java" in the conference INFORUM - Informatics Symposium 2021.

2021

MaxData Award - Informatics Excellence

I received the academic excellence award for the best student on the first year of the MSc on Informatics Engineering.

2018-2020

Diploma of Academic Merit

I received academic excellence awards for the three years of the Bachelor degree.

OTHER EDUCATION

Apr 2022	Science and the Media - bringing scientists, journalists and society cE3c, Faculdade de Ciencias da Universidade de Lisboa 
	2nd Grade on Concert Flute I had concert flute classes since the age of 6, and in 2014 got the 2nd grade of transverse flute by the Torres Vedras Conservatory.
	6th Grade on Classical Dance I attended classes of classical dance from 2006 to 2015, and achieved the 3rd Level of the "Certificate in Graded Examination in Dance" by Royal Academy of Dance, for the 6th grade of classical dance.

OTHER ACTIVITIES

2022, 2018	Monitor at Verão ULisboa and Ciências Open Day FACULDADE DE CIÊNCIAS DA UNIVERSIDADE DE LISBOA I helped in activities to promote Programming and Computer Science for high school students.
Jan 2019	Volunteer in the event Building the Future MICROSOFT AND IMATCH I participated as volunteer on the organization of the event Building the Future which joined technology, transformation and leadership to empower people and companies. The event was promoted by Microsoft in collaboration with imatch.
Jul 2017	Summer Course on Heritage and Creativity INSTITUTO POLITÉCNICO DE LEIRIA I participation on the International Summer Course about Heritage and Creativity.